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Bihar Engineering University, Patna

B.Tech 4th Semester Exam., 2025

Code: 105403		***Time: 03 Hours***
Semester: IV	***Subject: Operating System***	***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven questions only): **[2 x 7 = 14]**

(a) How many times will the string "OS" be printed?

```
#include <stdio.h>
#include <unistd.h>
main() {
   fork();
   fork();
   printf("OS\n");
}
```

(i) 2 (iii) 3
(ii) 4 (iv) 5

(b) Which scheduling algorithm may cause starvation?

(i) FCFS (iii) SJF
(ii) Round Robin (iv) FIFO

(c) A binary semaphore mutex is initialized to 1. Three processes P1, P2, and P3 request entry into the critical section in the order P1 → P3 → P2.
Each process executes:
wait(mutex);
critical section;
signal(mutex);
Which process will enter the critical section second?

(i) P1 (iii) P2
(ii) P3 (iv) Depends on scheduler

(d) A race condition occurs when:

(i) Processes execute sequentially (iii) CPU speed is low
(ii) Shared data is accessed concurrently (iv) Deadlock occurs

(e) External fragmentation occurs in:

(i) Paging (iii) Variable partition
(ii) Fixed partition (iv) Segmentation only

(f) Page size = 1 KB, logical address = 2500. Page number =?

(i) 1 (iii) 2
(ii) 3 (iv) 4

(g) Thrashing occurs due to _____

(i) High CPU speed (iii) Large memory
(ii) Disk failure (iv) Insufficient frames

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(h) Which disk scheduling algorithm moves head in one direction only?

(i) FCFS
(ii) SCAN
(iii) SSTF
(iv) C-SCAN

(i) Virtual machines mainly provide:

(i) Faster execution
(ii) Memory compaction
(iii) Hardware abstraction
(iv) Scheduling

(j) Dirty bit indicates:

(i) Page hit
(ii) Free frame
(iii) Modified page
(iv) Page fault

Q.2 Solve both questions : **[7 x 2 = 14]**

(a) Define system calls. Explain the types of system calls with examples. Describe how system calls provide communication between user programs and the OS kernel.

(b) Compare Monolithic and Microkernel Operating System architectures. Explain their working principles along with advantages and disadvantages.

Q.3 Solve both questions : **[7 x 2 = 14]**

(a) Define Process Control Block (PCB). Explain the contents of PCB and its role in process management and context switching.

(b) Consider the following processes:

Process	Arrival Time	Burst Time
P1	0	5
P2	1	4
P3	2	2
P4	3	1

Using Round Robin scheduling with a time quantum = 2 ms, draw the Gantt chart and find waiting time, turnaround time, and average waiting time.

Q.4 Solve both questions : **[7 x 2 = 14]**

(a) Define a thread. Explain the types of threads and compare them in terms of performance, flexibility, and overhead.

(b) Explain Peterson's solution for the critical section problem. Prove that it satisfies mutual exclusion, progress, and bounded waiting.

Q.5 Solve both questions : **[6+8=14]**

(a) A counting semaphore S is initialized to 3. Five processes execute the following operations in sequence:
P1: wait(S)
P2: wait(S)
P3: wait(S)
P4: wait(S)
P1: signal(S)
P5: wait(S)
Find the value of semaphore S after each operation. Identify the blocked processes.

(b) Explain any one classical IPC problem in detail:
(i) Readers–Writers Problem
(ii) Dining Philosophers Problem

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Q.6 Solve both questions : **[6+8=14]**

(a) Define deadlock in operating system. Explain the necessary conditions for deadlock?

(b) A system has 5 processes (P0–P4) and 3 resource types (A, B, C). The current resource status is given below:

Process	Allocation (A B C)	Maximum (A B C)	Available (A B C)
P0	0 1 0	7 5 3	3 3 2
P1	2 0 0	3 2 2	
P2	3 0 2	9 0 2	
P3	2 1 1	2 2 2	
P4	0 0 2	4 3 3	

(i) Calculate the Need matrix
(ii) Check whether the system is in a safe state
(iii) Find the safe sequence, if any

Q.7 Solve both questions : **[7 x 2 = 14]**

(a) Explain the principle of paging. Describe page allocation and address translation using page number and offset.

(b) For the page reference string
7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2
with 3 page frames, find the total number of page faults using the LRU page replacement algorithm.

Q.8 Solve both questions : **[6+8=14]**

(a) Explain file allocation methods: contiguous, linked, and indexed allocation.

(b) A disk has 200 cylinders (0–199). The disk request queue is:
98, 183, 37, 122, 14, 124, 65, 67
The disk head is initially at cylinder 53 and the direction of head movement is towards higher-numbered cylinders.
(i) Calculate the total head movement using FCFS
(ii) Calculate the total head movement using SSTF
(iii) Calculate the total head movement using SCAN
(iv) Calculate the total head movement using C-SCAN

Q.9 Write a short note on any four of the following: **[3½ x 4 = 14]**

(a) Logical address vs physical address

(b) Demand Paging

(c) Segmentation

(d) Deadlock Prevention

(e) Context Switching

(f) Device drivers

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